

Mapping Scalar Particle Preferences across Conditional Types in Czech: A CIT Analysis

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- **Minimizers/weak scalar particles** (e.g., *sebemenší* 'the slightest', English *even one*):
 - NPIs denoting a low degree on a scalar property Crnič (2011)
 - Licensed in downward-entailing/non-monotonic contexts:

- (1) a. John doesn't have *even one* book.
 b.??? John has *even one* book.

- **Maximizers/strong scalar particles** (e.g., *sebevětší* ‘the highest’, English *the most generous*):
 - Less studied; distribution not fully understood
 - Unlike English minimizers/maximizers, Czech offers fully productive pairs of them
 - in Czech they prefer to appear in concessive conditionals (even-if):

(2) a. Even if they offered *the most generous* terms, we'd refuse.

- **The Puzzle:** Czech *sebe-low* and *sebe-top* show strikingly divergent preferences across conditional types—challenging standard NPI theories (Barker 2018).

- **The Problem:** Czech NPIs (*sebe-low*: ‘the slightest’) and scalar particles (*sebe-top*: ‘the highest’) show **striking distributional preferences** across conditional types
 - *sebe-low* preferred in indicative & subjunctive conditionals (~87-98%)
 - *sebe-top* strongly preferred in even-if conditionals (~86%)
- **The Challenge:** Standard NPI licensing theories (Barker 2018) predict that NPIs should be licensed equally in all conditionals—but corpus data reveals systematic divergence
- **Our Approach:** Corpus study (n=500) + Conditional Inference Trees + semantic analysis via Crupi & Iacona (2022) to explain why particle selection tracks conditional type

- we focus on two kinds of Czech scalar particles:
1. NPIs of the type *sebe*-comparative denoting a low degree of a property (e.g., *sebemenší* ‘the slightest’ in (1) and (2), see also Dočekal and Juřen (2023)), referred to as *sebe-low*;

- (3) Budeme vděční, pokud přispějete
be.FUT.1PL grateful.M.PL if contribute.FUT.2PL
sebemenším finančním darem.
sebe-low.INS financial.INS gift.INS
'We will be grateful if you contribute the slightest financial
gift.' indicative conditional

- (4) Nacisté by s ním rychle
 Nazis.NOM would.SUBJ with him quickly
 zatočili, kdy-by pojali
 deal.PFV.3PL when-SUBJ.1PL take.PFV.3PL
 sebemenší podezření o jeho
 sebe-low.ACC suspicion.ACC about his.LOC
 stycích s námi.
 contacts.LOC with us.INSTR
 'If the Nazis had the slightest suspicion about his contacts
 with us, they would quickly deal with him.' subjunctive
 conditional

2. strong scalar particles of the type *sebe*-comparative denoting a high degree of a property (e.g., *sebevětší* 'the highest' in (3)), referred to as *sebe-top*;

- (5) I kdy-by měli sebevětší
even when-SUBJ.3PL have.PST.3PL sebe-top.ACC
příjem, bude jim to málo.
income.ACC be.3SG.FUT them.DAT it.NOM little
'Even if they had the highest income, it would be too little
for them.' even if conditional

- in our particular case, we focused on even if subjunctive conditionals in Czech;

- *Sebe-low* in Czech are NPIs with a likelihood profile corresponding to the English NPI *even one* (see Crnič (2011))
- intuitively, they seem to be preferred in the antecedents of indicative and subjunctive conditionals but not in the antecedents of *even if* subjunctive conditionals;
- such preferences are hard to capture in current theories of NPIs;
 - *sebe-low* are minimizers/NPIs, so they should be licensed in all kinds of conditionals (indicative, subjunctive, *even if*);

- our research goal is to map Czech NPIs' preferences in conditionals and aim at a theoretical explanation of the observed patterns;
- to achieve this goal, we conducted a corpus study (examples (1), (2), and (3) are taken from SketchEngine: Kilgarriff et al. (2014))
- the results were interpreted via non-parametric statistical recursive partitioning methods;
- finally, the consequences for the theories of conditionals are discussed.

- joint work with Jarmila Černá (Masaryk University)
- QR code of the slides:



- from SketchEngine corpus csTenTen23, we extracted conditionals with antecedents containing the NPI *sebe-low/top*
 - (CQL query: [lemma="jestli" | lemma="kdyby" | lemma="pokud"] []{1,5} [word="sebevětší" | lemma="sebemenší"], resulting in 1040 hits)
- we then obtained a random sample of 500 such sentences;
- the sentences were manually annotated for the type of conditional (indicative, counterfactual, *even if* contrafactual), the type of NPI (*sebe-low/sebe-top*), and the presence of local negation in the antecedent

- with all variables being of the boolean type;
- the descriptive results are shown in Figure 1.

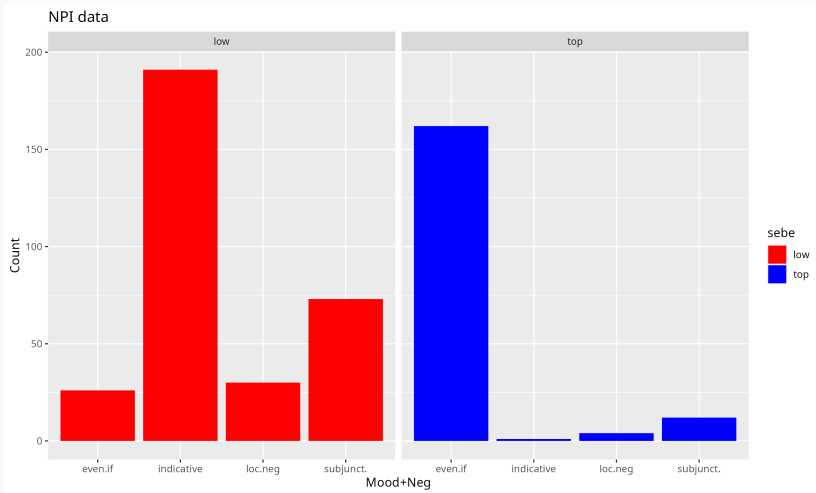


Figure 1: Figure 1: Descriptive results

- the inferential statistics were conducted using conditional inference trees (CIT, see Levshina (2020) a.o.), a non-parametric statistical recursive partitioning method;
- the analysis used the R (R Core Team (2024)) package *party* (Strobl et al. (2008));
- Figure 2 shows the individual tree fitted to the data:
- the dependent variable is the type of scalar particle (*sebe-low/sebe-top*)
- the independent variables are the type of conditional (indicative, counterfactual, *even if*) and the presence of local negation in the antecedent.

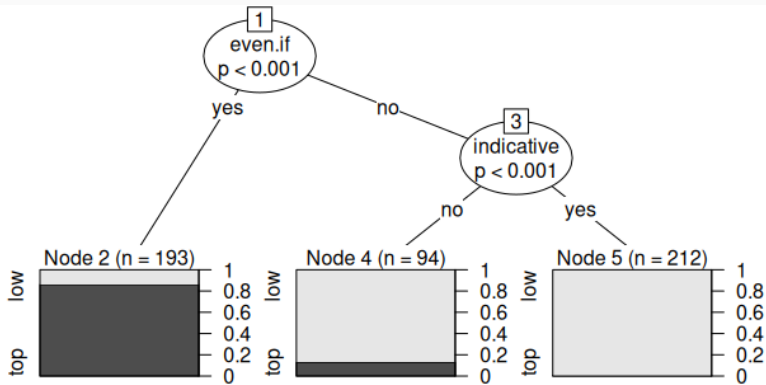


Figure 2: Figure 2: Conditional inference tree

- the top split (Node 1) is based on the variable *even if* contrafactual: the predictor (*even if* type of conditional vs. indicative+subjunctive) is the most important and significant ($p < 0.001$) predictor of the type of scalar particles;
- the data are then split into two subsets:
- the left subset (Node 2) contains sentences with conditionals of the type *even if*;
- the terminal Node 2 contains 193 sentences, with the majority (0.86) being *sebe*-top;

- the next split is based on the variable *indicative* (Node 3 – also significant): conditionals in the subjunctive mood (Node 4) contain 94 conditionals, with the majority (0.87) being *sebe-low*;
- the right subset (Node 5) contains 212 indicative conditionals, nearly all of which are *sebe-low* (0.98).

- the predictive power of the models is very satisfactory:
 - the CIT has a classification accuracy of 0.92 (with a baseline value of 0.51),
 - and the concordance index C is 0.94
- the results of the CIT were corroborated by a logistic regression model (not shown here – APPENDIX?) with the same dependent and independent variables, yielding very similar results.

Interpretation of the results

- we assume that all types of conditionals (indicative, subjunctive, *even if* contrafactual) are non-monotonic (solution of most problems of conditionals in natural language)
- and adopt the most standard semantics of conditionals: Stalnaker (1976), Lewis (2013)

- for licensing of NPIs, we follow the scope theory of NPIs licensing by Barker (2018), which can account for NPI licensing in non-monotonic contexts;
- NPIs are licensed if their wide scope doesn't entail their narrow scope:
 - licensing by negation
 - the NPI signals the narrow scope of the indefinite w.r.t. negation

(6) John didn't see *any* students.

- $\exists x[student'(x) \& \neg see'(J, x)]$
- $\not\models \neg \exists x[student'(x) \& see'(J, x)]$

- in non-monotonic contexts, the entailment does not hold, so NPIs can be licensed too

(7) Exactly one student passed any exam.

a. $\exists y [\text{exam}(y) \wedge \exists! x (\text{student}(x) \wedge \text{passed}(x, y))] \not\models$

b. $\exists! x [\text{student}(x) \wedge \exists y (\text{exam}(y) \wedge \text{passed}(x, y))]$

- countermodel: Alice passed E_1 only; Bob passed E_2 only
- wide scope of the indefinite is TRUE: E_1 was passed by exactly one student (Alice)
- narrow scope of the indefinite is FALSE: no exam was passed by exactly one student
- for Barker, the NPI is licensed since the wide scope does not entail the narrow scope;

Indicative conditionals

- the examples of the type (1) with *sebe*-low in the antecedent of indicative conditionals can be analyzed as follows:

- (8) If you contribute even the slightest financial gift, we will be grateful.
- you contribute even the slightest financial gift ... p
 - we will be grateful ... q
 - $p > q \approx \Box(p \rightarrow q)$ (Stalnaker)
 - $\forall w' \in C$: if p is true at w' , then q is true at w'
- strict conditional: universal quantification over a fixed modal base (the context set C of epistemically accessible worlds), intersected with p-worlds

- it is non-monotonic (antecedent strengthening fails);
- the antecedent p is non-monotonic, so the NPI can be licensed via Barker's scope theory;
- Non-monotonicity + additivity is too weak to explain the minimizer/maximizer split. Following monotonic approaches to conditionals (e.g., Alxatib on *only if*), we need stronger degree-based implicatures to predict why *sebe-low* dominates here while *sebe-top* is dispreferred.
 - to be worked out

- in Barker's theory, it is similar to the licensing of NPIs in sentences with universal quantifiers: also only in the restrictor

- (9) a. Every student with any experience in programming passed the exam.

$$(i) \quad \exists e \left(E(e) \wedge \forall x ((S(x) \wedge H(x, e)) \rightarrow P(x)) \right) \not\vdash \\ \forall x ((S(x) \wedge \exists e (E(e) \wedge H(x, e))) \rightarrow P(x))$$

- b. #Every student with a programming experience passed any exam.

- as for universally quantified sentences, for indicative conditionals, the wide scope of the indefinite NPIs fails to entail their narrow scope due to non-monotonicity of the conditionals;

- (10) If the Nazis had even the slightest suspicion, they would quickly deal with him.
- a. the Nazis had even the slightest suspicion ... p
 - b. they would quickly deal with him ... q
 - c. $p \Box \rightarrow q$
 - d. $\forall w' \in f(w, \llbracket p \rrbracket) : q$ is true at w'
- again the standard analysis after Lewis/Stalnaker:
 - variably strict conditional: universal quantification over a variable modal base determined by a similarity ordering that depends on p ;

(11) $\forall w' \in f(w, \llbracket p \rrbracket) : q$ is true at w' (repeated (10-d))

- the selection function f returns the closest p -worlds to the evaluation world w : which worlds count as “closest” varies with p ;
- again non-monotonic, so the NPI can be licensed via Barker’s scope theory;

Even if counterfactual conditionals

- *even-if* conditionals are known to come in two flavors (Pollock (1976), Bennett (1982), Lycan (1991)):

- (12) a. Even if they were to raise my salary double, I would
quit the job. introduced-if
b. Even if John said one word, I would leave. standing-if

- focusing on introduced-if: it implicates that the speaker will quit the job regardless of the circumstances – the speaker is committed to the truth of the consequent – Guerzoni and Lim (2007) calls this inference “factivity”;

(13) Even if John said one word, I would leave. standing-if
(repeated (12-b))

- standing-if conditionals are not 'factive': can be true (like material implication) if the consequent is false (forcing falsity of the antecedent)

- Czech “sebe-low” seem to appear (more empirical work needed) mostly/only? in standing-if conditionals, which lack the factivity implicature: (14)
- and they are much less frequent in *even if* counterfactuals than in indicative and subjunctive conditionals (see Figure 1 and CIT results)

(14) I kdyby ses toho obrazu
 even if.3SG you.GEN this.GEN picture.GEN
 dotkl se bemeně, okamžitě
 touch.PFV.SBJV.2SG sebe-low.INST immediately
 se spustí alarm.
 REFL start.FUT alarm.NOM
 ‘Even if you were to touch the picture the slightest, the
 alarm would go on immediately.’ standing-if

I'm not sure whether it is appropriate to give a uniform analysis of both types of even-if conditionals (some like Crupi and Iacona (2022) analyse some formally marked *even if* conditionals as regular conditionals)

- but be it as it may, there is some conditional core in both types of *even if* conditionals that is non-monotonic;
- that itself should be sufficient for licensing NPIs via Barker's scope theory;
- however, Barker's theory fails to explain the observed dispreference of *sebe-low* in *even if* conditionals;
 - and it is silent about the preference for *sebe-top* in such conditionals: it is a theory of NPI licensing, not of scalar particles in general;

- let's add to Barker's theory a pragmatic component: Crnič (2011) (general framework) and Dočekal and Juřen (2023) (for Czech) in explaining the NPI *sebe*-low as a scalar particle with unlikelihood presupposition having wide scope:

(15) [even]($\phi > \psi$) ($p \triangleleft_C q$: the prejacent p is at most as likely as focus induced alternatives q in C).

- the prejacent is either as likely (non-monotonic analysis) or less likely than the alternatives: likelihood translates into entailment – $p \wedge q$ entails both p and q and is less likely than either of them;

- covert *even* is best understood as a formalization of the likelihood/normality component inherent to overt *even*, not an independent pragmatic mechanism
- Maximizers first: A complete account must model *sebe-top* distribution before *sebe-low*. The unlikelihood presupposition alone is insufficient: superlatives are fully licensed in episodic contexts, so the preference/dispreference must emerge from interaction with conditional structure

- Crnič's ordering is completely different from the standard ordering of worlds in conditionals (similarity-based in counterfactuals, epistemic in indicatives) and it works in extensional contexts too:

(16) Petr nepřispěl sebemenší
 Peter.NOM not.contribute.PST.3SG sebe-low.INST
 částkou.
 amount.INST
 'Peter didn't contribute even the slightest amount.'

- not contributing the slightest amount entails not contributing any amount at all (all the contextual alternatives) and therefore is less likely than contributing any amount;

- two orthogonal orderings:
 - the standard ordering of worlds in conditionals (similarity-based/epistemic) for the conditional semantics;
 - the likelihood ordering for the scalar particle semantics;
 - but they seem to converge for indicative and subjunctive conditionals as tested with *sebe-low*;
- however, they diverge for *even if* conditionals;

Finding the right analysis of *even if* conditionals is a task for future research

- but it seems that Crupi and Iacona (2022) works a bit better than some other approaches
- some cross-linguistic remarks and some speculations about Crupi and Iacona (2022)'s predictions

English

- (17) a. We'll be grateful if you contribute any money.
indicative
(i) ??We'll be grateful if you contribute the greatest amount.

- (18) a. If they had the slightest suspicion, they would have acted. counterfactual
(i) ?If they had the greatest suspicion, they would have acted.
- b. Even if they had the greatest income, it would be too little. even if
(i) ??Even if they had the slightest income, it would be too little.
- and probably similar patterns can be observed in other languages (German, ...)

Why Maximal Endpoints Dominate Concessives

Crupi & Iacona (2022) analyze concessives as **no-matter whether p** (not their notation):

(19) $(p > q) \wedge (\neg p \Box \rightarrow q)$

(20) Even if it were to rain, we will play the football match.

- a. p : it rains
- b. q : we will play the football match
- c. $(p > q)$: in the epistemic worlds of p , q holds
- d. $(\neg p \Box \rightarrow q)$: in the closest worlds where $\neg p$ holds, q also holds

- plus their Observation 3: $(p > q) \wedge (\neg p \Box \rightarrow q)$ involves an asymmetry between p and $\neg p$, where “the connection between $\neg p$ and q is more natural, or less surprising, than the connection between p and q ”

(21) Even if it were to rain, we will play the football match.

- Here, the natural expectation is: good weather $\rightarrow$ going out. The concessive flips this: it asserts that even the unnatural case (bad weather) still yields the consequent.
- the asymmetry is derivable from their formalization;

- back to our case:

(22) Even if they received the highest income, it would be too little for them.

- a. p : they receive the highest income
- b. q : it would be too little for them
- c. $(p > q)$: in the closest epistemically accessible p -worlds, q holds
- d. $(\neg p \Box \rightarrow q)$: in the closest worlds where $\neg p$ holds, q also holds

- I'm simplifying Crupi & Iacona's evidential conditional here and representing it as a standard counterfactual $\neg p \Box \rightarrow q$; strictly speaking, their operator is stronger (it also builds in a support/incompatibility condition).

Why sebe-low Fails in Concessives

- Crupi & Iacona's concessive: $(p > q) \wedge (\neg p \Box \rightarrow q)$
- $\neg p \Box \rightarrow q$ encodes the **stereotypical connection** (closest worlds)
- With **sebe-low** ($p = \text{minimal degree}$): $\neg p = \text{average/high degree}$
 - Normal expectation: average/high income $\rightarrow$ no quarrel ($\neg p \Box \rightarrow \neg q$)
 - Concessive requires $\neg p \Box \rightarrow q \rightarrow$ **clashes with normality** $\rightarrow$ pragmatically infelicitous
- With **sebe-top** ($p = \text{maximal degree}$): $\neg p = \text{average/low degree}$
 - Normal expectation: low income $\rightarrow$ quarrel ($\neg p \Box \rightarrow q$) $\rightarrow$ **perfect fit**
- Thus, sebe-top is selected because only maximal endpoints preserve the required $p/\neg p$ normality asymmetry.

Why sebe-top Dominates Concessives

- the natural expectation is: receiving the slightest income $\rightarrow$ it would be too little for them
- **sebe-top** denotes an ‘evidential outlier’—the most extreme degree—thereby **maximizing the contrast** between the stereotypical expectation ($\neg p \square \rightarrow q$) and the asserted invariance ($p > q$)
- **Why sebe-low is pragmatically weak::** If q holds even for the *smallest* p , it doesn’t necessarily violate the expectation that q might fail for a *large* p
- Therefore, **sebe-top** is the superior pragmatic fit for communicating “no-matter-what” concessives

- This follows directly from Crupi & Iacona's formalism
 - In $(p > q) \wedge (\neg p \square \rightarrow q)$, the stereotypical connection is $\neg p \rightarrow q$
 - Using **sebe-low** (minimal) in p creates *weak* violation $\rightarrow$ poor fit
 - Using **sebe-top** (maximal) in p creates *strong* violation $\rightarrow$ optimal fit

Theoretical Consequences (support for sem. appr. to even-if)

- Rules out: Treating even-if as ordinary conditionals + pragmatic component (Guerzoni & Lim 2007, Bennett 1982)
- Supports: Distinct conditional semantics with built-in asymmetry between p and $\neg p$ (Crupi & Iacona 2022, Lycan 1991)
- If even-if were just [conditional + scalar operator], we'd expect **sebe-low** (NPI minimizer) to be licensed equally in all conditionals
- But the corpus shows systematic **sebe-top** preference in even-if conditionals (86%)
 - predicted by Crupi & Iacona's support asymmetry

Summary

- the current theories of NPIs (Barker (2018), Crnič (2011)) can explain the licensing of weak scalar particles (NPIs) in indicative and subjunctive conditionals but fail to explain their dispreference in concessive conditionals;
- the strong scalar particles (maximizers) are strongly preferred in concessive conditionals but are dispreferred in indicative and subjunctive conditionals, which is also not predicted by the current theories;
 - not their fault since we don't have a real theory of these items yet;

- first promising step: follow the suggestion of Crupi and lacona (2022) and analyze concessives as involving an asymmetry between p and $\neg p$ (stereotypical connection – the normal worlds) that is not present in other conditionals;

Thank you for your attention!

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